GlycoTwin

The Virtual Biomarker Lab for Type 2 Diabetes

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Happiest Health Digital Twin Challenge 2026

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Problem statement

The gap

Type 2 diabetes is monitored through laboratory HbA1c tests every 3–6 months. Between visits, clinicians cannot see how a patient's **daily activity, sleep, and heart-rate behaviour** relate to glycaemic trajectory.

Clinical data — accurate but sparse

- HbA1c, fasting glucose, BMI
- Diagnosis history
- A few points per year

Wearable-like data — dense but indirect

- Steps, active minutes
- Resting heart rate
- Sleep duration & consistency

No unified, explorable view of both — and no safe way to ask "what if?" without over-claiming.

Healthcare use case: clinician-facing decision support

1. Select a **synthetic patient**; review historical clinical records and the **last laboratory date**.
2. Inspect **14-day wearable trends** (steps, activity, resting heart rate, sleep).
3. Run **model inference** for an estimated future HbA1c with an **exact per-feature explanation**.
4. Explore **what-if scenarios**: change lifestyle inputs, compare baseline vs scenario estimates.
5. Review **configurable follow-up alerts** that prompt *"consider clinical/laboratory review"*.

Intended outcome: a transparent decision-support concept that unifies historical clinical and dynamic wearable data — while stating on every screen that predictions are *not* laboratory results and never replace clinical judgement.

Out of scope: diagnosis, medication guidance, risk probabilities, real device ingestion, clinic/booking integration.

Data: fully synthetic, patient-linked, documented

Committed sample (seed 20261012)

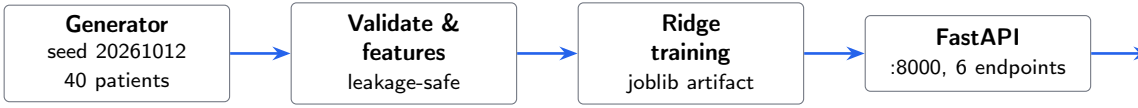
- 40 synthetic patients, generated locally
- 3 dated clinical observations per patient (HbA1c, glucose, BMI)
- 14 daily wearable-like records per patient
- Future target: synthetic HbA1c 90 days after cut-off
- Provenance, formula, and schema in `docs/`

Honesty rules

- No real people or measurements
- Target built by a documented generator formula
- Labelled *synthetic* in every response
- No merging of unrelated datasets

Leakage discipline: features use only observations $\leq$ prediction time — latest eligible clinical record and exactly 14 wearable days ending at the cut-off; splits are **patient-disjoint** 70/15/15 (tested).

Architecture: end-to-end implemented pipeline



Data layer
CSV in → validated tables, split files

ML layer
scaler + model saved together, loaded per request

Presentation layer
FastAPI + React; `/api` dev proxy; CORS restricted

No database, cloud, or hardware required — one command per component.

AI/ML approach and evaluation design

Task: regression — estimate synthetic future HbA1c (90-day horizon) from clinical history + 14-day wearable aggregates.

Two deterministic scikit-learn candidates, fitted on train only:

- **Clinical-only baseline** — age, diabetes history, latest HbA1c, glucose, BMI
- **Clinical + wearable** — plus 14-day means of steps, activity, resting heart rate, sleep duration, sleep consistency

Design guarantees

- Selection by **validation MAE**; test split held out entirely
- **Patient-disjoint** splits — no identity leakage (tested)
- Baseline vs combined comparison as required by the brief

Measured results (synthetic data only)

Candidate	Val MAE	Test MAE	Test RMSE
Clinical-only Ridge	0.231	0.149	0.174
Clinical + wearable Ridge (selected)	**0.206**	**0.142**	**0.159**

Unrounded values in `models/synthetic_hba1c_metrics.json`; selected by validation MAE, evaluated on 6 held-out synthetic patients.

What these numbers do *not* mean

The target is generated by a known formula; recovering it demonstrates a *working, leakage-safe pipeline*, not clinical accuracy. Six test patients cannot establish real-world performance. No uncertainty, no risk probability, no causal claims.

Explainability: exact, tested, honest

How it works

- Model is `StandardScaler` + `Ridge`
- Each feature's contribution = coefficient × scaled value
- + intercept reconstructs the estimate **exactly**
- Shown in the dashboard as a contribution chart
- **Verified by an automated test** (sum = prediction)

SHAP decision

SHAP was considered and **deliberately not added**: for a linear model SHAP's `LinearExplainer` returns the same attributions, with a heavy dependency chain in return.

"`Contributions explain the model's arithmetic, not clinical causation.`"

Inference API (FastAPI)

Endpoints

- `GET /health` `GET /patients` `GET /patients/{id}`
- `POST /predict` `POST /simulate` `GET /alerts`

Guarantees

- Real artifact loaded — **never hard-coded**
- Pydantic validation; explicit 404 / 422 / 503
- Uncertainty field present but `null`
- `deterioration_risk_available: false` everywhere

Demo alert rule

Latest synthetic HbA1c $\geq$ **8.0%** $\rightarrow$ flag for *consider clinical/laboratory review*.

Scenario engine

Re-runs the same pipeline on modified inputs; returns baseline, scenario, and the difference — labelled *exploratory, not a treatment outcome*.

Clinician dashboard (React + TypeScript)

Panels

- Patient selector + API health badge
- Clinical summary with **last lab date**
- Five wearable trend charts with units
- Prediction card + contribution chart
- What-if simulator (5 sliders, observed vs scenario table)
- Alert panel with *Review patient*

Safety engineering

- Persistent synthetic/prototype banner
- Loading / empty / error states everywhere
- Results **cleared before each request** — failures never show stale numbers
- Inference runs only on explicit user action
- Units & dates on every axis

Stack: React 19, Vite, TypeScript (strict), Tailwind 4, Recharts — connected through a same-origin `/api` proxy; CORS restricted to the local origin.

Engineering quality: verified, not claimed

15/15 automated tests passing (pytest)

- Deterministic generation; schema/range/linkage validation
- Feature-window leakage protection; patient-disjoint splits
- Training reproducibility and held-out evaluation
- API behaviour: browsing, prediction, simulation, alerts
- **Explanation exactness:** contributions + intercept = prediction
- **CORS:** local origin allowed, foreign origins refused

Frontend checks: strict `tsc` typecheck + Vite production build; live HTTP verification of every request the UI performs.

Reproduce: `python -m pytest` · `npm run build` — both documented in the README.

Limitations and responsible use

Stated openly in the README, model card, API, and every UI screen

- **All data is synthetic and project-generated** — no clinical, causal, prevalence, or real-world claims
- The target comes from a documented generator formula — the model learns synthetic relationships
- Evaluated on **6 held-out synthetic patients** — not a stable performance estimate
- **No uncertainty quantification**; no deterioration-risk classifier (no valid future-event label exists)
- Alert threshold is a **configurable demo rule**, not clinically approved
- Not a medical device, not a diagnosis, not a substitute for lab testing or clinician judgement

Project instructions required honesty over impressive numbers: we report only measured results, and a doc section records *why* SHAP was not used.

Outcome: what works today

Delivered (Phases 1–5 complete)

- Reproducible data → features → model → API → UI chain
- Live dashboard consuming real API responses
- Real-pipeline prediction with exact explanations
- Scenario simulator and configurable alert rule
- 15 passing tests; full setup documented in README

Proof points for the challenge

- Historical clinical + dynamic wearable inputs demonstrably unified
- Output generated by actual code, evaluated honestly
- Dashboard consumes backend outputs end-to-end
- Open-source (MIT), public GitHub repository
- Runnable locally in three terminals

A small, working, honest system — before complexity.

Deliverables & links

Repository (public)	`github.com/Ankush443/GlycoTwin`
Licence	MIT (open source)
Architecture diagram	`docs/Architecture_Diagram.pdf`
Presentation	`docs/Presentation.pdf`
Demo video (20+ min)	recorded walkthrough of problem, data, pipeline, evaluation, live demo, limitations

Thank you — GlycoTwin: synthetic digital-twin insight, clinical honesty first.

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